The Special Theory of Computational Relativity (CoReX): An Exposition and Critical Evaluation

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Abstract

CoReX is an interactive monograph, published as a single web page, that argues computation and physics are two readings of one geometric object: a zero-point seed, A_0, acted on by the rotation algebra so(N). From that one mechanism it claims to resolve P versus NP, the double-slit puzzle, the three-body problem and all seven Clay Millennium Problems. In this dissertation I rebuild the theory from the page alone (its text, its equations and its interactive demonstrations) and then test it against standard mathematics and physics.

The theory rests on a single postulated axiom, Ω_0, and on one device that recurs everywhere: the "nodal firewall", a surface where two paths meet in anti-phase, so that e^{iπ} + 1 = 0 forces the amplitude to zero. The author grades the claims candidly. One is a conditional in-manifold theorem (P = NP), five are structural analogies, and one is historical commentary (Poincaré).

My conclusion is that the framework holds together as an axiom system but does not connect to established results. Its central object, $\lim_{x\to 0} 0/x = \lambda > 0$, equals 0 in the real numbers, and the transfer principle it appeals to preserves that fact. Several derived claims fail simple checks: the Fourier "A_0 kernel", the Navier–Stokes argument from $\det(CM_5) = 0$, and polynomial-time search by continuous embedding. What remains is a consistent visual and notational language, and some clear statements of falsifiability that the author supplies.

Keywords: computational relativity, Euler identity, so(N), P versus NP, Gibbs phenomenon, nodal firewall, Landauer bound.

1. Introduction

This dissertation studies one object: the interactive monograph *The Special Theory of Computational Relativity* (CoReX v3.1, Pierre LeBaux), published as a single web page with seven chapters, fifteen animated canvases and a short appendix. Everything I say about the theory comes from that page. I use no outside literature to describe it; standard mathematics enters only in Chapter 8, as the yardstick.

1.1 The claim in one paragraph

The motto, *Veritas et Ratio Invariantia Sunt in Toposo* $*\mathbb{V}$, says that truth and reason are invariant in a hyperreal topos. The working claim behind it is that every physical field, state space and conservation law is a projection of a zero-point scalar seed A_0 under so(N), the Lie algebra of rotations. The equals sign is reinterpreted as a G-equivariant "transformer" between two faces of one orbit, which is how the page can write P = NP ⇔ E = mc² as a single operator statement.

1.2 Method and scope

- I take every definition, number and diagram from the page. Where the page is silent, I say so.
- Chapters 2–7 set out the theory as its author does; Chapter 8 tests it.
- The page grades each Millennium reading: Grade C (conditional in-manifold theorem), Grade S (structural analogy) or Grade H (historical commentary). I keep those labels and check whether the rest of the page respects them.

1.3 The page's own honesty clauses

The monograph is frank about its status. It says that Ω_0 "is postulated, not derived", that the $O(N^2)$ bound is the cost of a single step and not a proof of convergence, and that the unrelativized Clay questions are "not claimed resolved". These statements set a fair standard: the theory should be judged as a self-consistent axiom system and as an analogy, not as a proof of any open problem.

1.4 Structure

Chapter 2 sets out the axioms and Chapter 3 the P = NP argument. Chapter 4 covers the seven Millennium readings, Chapter 5 the four experiments (laboratory, double slit, three bodies, time machine) and Chapter 6 the Fourier analysis. Chapter 7 describes the two applications, Chapter 8 evaluates everything, and Chapter 9 concludes.

2. Foundations: the root axiom and the seed

Everything in the theory follows from one postulated statement, Ω_0, which the page calls the "Master Root Axiom" or "Equivariant Void". What comes later is presented as what Ω_0 entails, not as separate results.

2.1 The root axiom

```
\Omega_0:\; \mathcal{M}_N(A_0) = \mathrm{Proj}_{\mathfrak{so}(N)}(A_0),\quad
\forall T\in\mathcal{T}:\; T(\mathrm{Veritas})=\mathrm{Veritas}\wedge
T(\mathrm{Ratio})=\mathrm{Ratio},\quad e^{i\pi}+1=0 \Rightarrow
|\psi_{\mathrm{node}}|=0
```

In words: the manifold of states is the so(N) orbit of the seed; truth and reason are unchanged by every admissible transformation T; and a half-turn of phase cancels the amplitude at a node.

2.2 The seed A_0

The seed is defined by a limit, which the page calls the unique fixed point of the full symmetry group:

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A_0 \equiv \lim_{x\to 0}\frac{0}{x} = \lambda \in {}^{*}\mathbb{R}^N
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The page defends this by saying it never divides by zero, only by x while x tends to zero, so the limit is finite. The seed lives in a hyperreal topos, *𝕍 = V^I/🚫, and Łoś's transfer theorem is cited to carry statements between the ordinary and hyperreal levels. The zero point is "the only locus where every generator of the algebra vanishes at once", and everything later (the zeros of ξ, mass gaps, homotopy spheres) is said to inherit its stability from there.

2.3 The rotation algebra

A state evolves as $\mathcal{M}_n(A_0) = \{ U(t)\cdot A_0\cdot U^\dagger(t) \}$, driven by a skew-symmetric generator $\Omega(t) = \Sigma\, c_a{}^b(t)\, J_a{}^b$ in so(N). The number of independent rotation planes is

```
R_N=\binom{N}{2}=\frac{N(N-1)}{2}
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This count grows quadratically in N, and it is the quantity that later replaces the exponential 2^N of a decision tree. The laboratory chapter lets the reader vary N from 2 to 10 and watch R_N grow while the seed stays fixed at the origin.

2.4 The equal sign as a transformer

The page's first postulate reads "=" as a G-equivariant map between two faces of one orbit, not as identity. Since invariants survive projection, it treats conservation of mass–energy and collapse of the state space as two readings of one operator equation, written $P = NP \Leftrightarrow E = mc^2$.

2.5 Euler's identity as phase signature

A half-turn in any SO(2) plane sends the identity state back to zero. The amplitude formula used throughout is

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|\psi_{\mathrm{node}}|=\tfrac12\,|\psi\,(1+e^{i\pi})|=0
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Any branch that picks up a relative phase $\Delta\theta = \pi$ against the ground state is cancelled. The page calls the surface where this happens a **nodal firewall**.

2.6 Why "special"

The FAQ explains the word as in special relativity: the computational metric is fixed and flat, and complexity is measured against a rigid grid of "computational light-cones" anchored at A_0. A general theory, in which the metric curves under problem load, is left for future work.

2.7 Axioms versus postulates

The appendix separates three axioms (the seed, the hyperreal topos with transfer, and invariance of c and rest energy) from eight postulates: Euler annihilation, so(N) embedding of complexity, Cayley–Menger volume invariance, a multi-lobe orbital hierarchy, zero entropic friction, Dirichlet phase filtering, causal sector isolation and the Landauer bound. The full table is in the Appendix.

3. The constructive reading of P = NP

The monograph claims that the exponential decision tree of a Boolean problem never has to be searched, because it embeds in the continuous rotation geometry of so(N), where wrong branches cancel themselves. The page labels this Grade C: a conditional theorem, valid inside the Ω_0 axiom system.

3.1 Statement

Let Φ be a Boolean formula in N variables. The decision tree over {0,1}^N is mapped to trajectories U(t) in SO(N), driven by skew-symmetric generators $J_a{}^b = e_a e^{bT} - e^b e_a{}^T$. Updating each of the R_N rotation variables costs O(1), so

```
T_{\mathrm{solve}}(N)=R_N\cdot O(1)=\frac{N(N-1)}{2}\cdot O(1)=O(N^2)
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3.2 The mechanism

The page's axiomatic chain has four steps:

1. State spaces are so(N) orbits of the seed (Ω_0).
2. The cube {−1,+1}^N embeds injectively in the sphere S^{N−1}, so search becomes geodesic flow in the R_N rotation planes.
3. Along the flow dU/dt = Ω(t)U(t), non-optimal paths pick up a relative phase Δθ = π, and by Euler's identity their amplitude vanishes at a nodal firewall.
4. The system follows the unique optimal geodesic in O(N²) time.

3.3 The demonstration

The first canvas under "Seven Problems" draws a binary tree with eight leaves, labelled 2^3, whose branches fade as an annihilation wave passes through. A circle of three rotation lines ($R_3 = 3$) sharpens in step, and a gold pulse runs from the tree to the circle. The caption ends with "✓ collapsed" when the counter reaches 3. In miniature it shows the whole argument: exponential branching on the left, quadratic rotation planes on the right.

3.4 What the page concedes

Two caveats are printed next to the theorem. First, O(N²) is "the per-step cost of driving the flow"; that the flow reaches a stable integer vertex in polynomial time is "the framework's postulated convergence layer". Second, the result is an in-manifold theorem under Ω_0, and the unrelativized Clay question is not claimed resolved. So the actual statement is conditional: *if* the Ω_0 flow converges in polynomially many steps, *then* satisfiability costs O(N²) updates. All the difficulty sits in the "if", which is where Chapter 8 spends its time.

4. Seven Millennium Problems, one mechanism

The monograph reads each Clay problem as a lower-dimensional projection of the same invariants: the seed A_0, the rotation planes of so(N) and the anti-phase firewall $|\psi| = 0$. Only one reading (P versus NP) is graded as a theorem, and then only conditionally. Five are graded as structural analogies, and one as commentary on an already proved result.

#	Problem	Reading on the page	Grade	Caveat printed by the page
I	P versus NP	$R_N = C(N,2) \Rightarrow T(N) = O(N^2)$; tree embeds in rotation planes	C	Conditional on the Ω_0 convergence layer
II	Riemann Hypothesis	$\xi(s) = \xi(1-s)$ is a mirror across the A_0 axis; off-axis zeros have density forced to 0	S	Symmetry reading, not an analytic proof
III	Yang–Mills mass gap	$E_0 = \lambda c^2 > 0$ gives a positive floor; $\Delta M = \hbar\omega_{a_0}/c^2 > 0$	S	Gap imported from the seed's positivity; lattice QCD not re-proved
IV	Navier–Stokes smoothness	$\det(CM_5) = 0$ blocks rank inflation; anti-phase cancels vorticity blow-up	S	None printed
V	Birch & Swinnerton-Dyer	Each rational generator of $E(\mathbb{Q})$ is one rotation plane; ord $L(E,s)$ at $s = 1$ counts the same planes	S	None printed
VI	Hodge conjecture	Hodge classes are invariant subobjects of a topos GUT-Cat under classifier Ω_{a_0}	S	None printed
VII	Poincaré conjecture	Ricci flow neck-pinches annihilated at firewalls; M^3 deforms to S^3	H	Proved by Perelman (2003); CoReX re-derives the mechanism, not the credit

4.1 The common template

All seven chains have the same three links. First, a structure from the problem is assigned to the seed's geometry: the zeros of ξ to a mirror axis, a spectrum to a vacuum floor, a flow to a trajectory in so(N). Second, the pathological case (an off-axis zero, a vanishing gap, a blow-up, a neck-pinch) is declared anti-phase to the ground state, $\Delta\varphi = \pi$. Third, $e^{i\pi} + 1 = 0$ annihilates it, and the desired statement is all that is left. In each case the conclusion is stated as a corollary of Ω_0.

4.2 The seven canvases

Each reading has an animation. In II, off-axis zeros are pushed back to Re(s) = 1/2 and ring as they lock in. In III, excitations rise from a vacuum floor and are absorbed at a UV cutoff line. In IV, concentric vortex rings tighten and a red annulus of anti-phase flares before the core can diverge. In V, four gold "generator" points ride on the curve and light bars labelled ord 1 to ord 4. In VI, a wandering phantom class snaps onto an invariant rational cycle. In VII, a wobbly loop sprouts a neck-pinch that is annihilated, leaving a smooth circle labelled "S^3 – flow complete". The animations illustrate the story; they do not compute the quantities they name.

4.3 Reading the grades

The grading is the most useful thing on the page. It tells the reader which statements are mathematics (the $O(N^2)$ cost of a defined flow), which are analogy (the mirror axis for ξ) and which are history (Perelman). Chapter 8 asks whether the FAQ, which at one point says that, inside Ω_0, the seven readings are "theorems of the axiom system", stays consistent with those grades.

5. Four experiments

Chapter IV of the monograph turns the axioms into four interactive experiments. Each has a control panel, a canvas and a live "State" readout, and all four make the same move: a pathological branch is labelled anti-phase and removed at a firewall.

5.1 Laboratory

The laboratory has two sliders (dimension N from 2 to 10, rotation speed from 0 to 2) and three toggles (planes, nodes, orbits). Lines through the origin stand for the R_N = N(N−1)/2 rotation planes, concentric circles carry N orbiting points, and the seed A_0 sits

at the centre. For N = 5 there are 10 planes; for N = 10 there are 45. The experiment is a picture of the counting formula and claims nothing beyond it.

5.2 Double slit

The page's reading drops both wave–particle duality and the collapse postulate. A classical particle passing through the slits excites the vacuum field of A_0. Where the two path phases agree, the field reinforces and the particle lands; where they are anti-aligned, e^{iπ} + 1 = 0 and the particle never lands. The dark bands are nodal firewalls. The canvas draws two slits (A in green, B in red) and a screen whose brightness follows

```
I(y)=\cos^2\!\left(\frac{k\,\Delta r}{2}\right),\qquad k=\frac{2\pi}{\lambda}
```

where Δr is the path difference from the two slits. The "observer" toggle switches to which-path measurement: wavefronts and firewall lines disappear, dashed straight paths take their place, and the screen shows two smooth bands. The FAQ describes the framework as observer-free. Recording a path is not a collapse but a firewall event, with the observer acting as a boundary condition.

5.3 Three-body problem

The page asserts that the phase seed $A_0 = \lambda > 0$ locks three bodies onto the figure-eight choreography, with phase offsets of exactly 2π/3. Trajectories that leave the eight are anti-phase against the ground state ($\Delta\varphi = \pi$) and meet a nodal firewall at the crossing node. The canvas shows the eight with three bodies, optional trails, and three red "chaos twins" that start 0.004 away and drift outward until a pulse at the crossing re-seeds them. The "Perturb twins" button multiplies the offset. The slogan is that chaos is not integrated, it is forbidden.

5.4 Time machine

Here, reversing time flips the orientation of a trajectory (L_mesh → −L_mesh) but leaves the rest energy $E = mc^2$ unchanged. A conveyor belt, a carriage and a clock hand run backwards while an energy badge stays fixed. A vertical firewall separates PAST from FUTURE. With the firewall on, the carriage stops at the boundary and the label reads "crossing forbidden"; with it off, the sectors connect and the carriage crosses. The page also adds a Landauer meter with the bound

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\Delta E \ge k_B T \ln 2
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and states an impossibility result: macroscopic time travel is excluded because reversing a macroscopic history makes the dissipated heat diverge. Cheap orientation inversion is set against expensive history erasure. So the experiment describes a time machine that cannot be built, and the page presents exactly that as its content: orientation inversion, energy conservation and boundary isolation, "nothing more".

6. Fourier analysis without the Gibbs overshoot

The fifth experiment claims that a phase-modified kernel removes the 8.95% Gibbs overshoot at a jump without the smearing that other filters bring. Of all the chapters, this is the only one whose claim can be checked numerically on the page itself.

6.1 The classical problem

A Fourier partial sum of a step function overshoots the step by about 8.95% of the jump height, however many harmonics N are used. The page writes the constant as

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\frac{1}{\pi}\,\mathrm{Si}(\pi)-\frac12\approx 0.0895
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and calls it the Wilbraham–Gibbs phenomenon. In the monograph's reading, the ringing is the visible signature of the origin singularity, concentrated exactly at the point A_0.

6.2 The A_0 kernel

The page replaces the Dirichlet kernel $D_n(x)$ with

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D^{*}_n(x)=D_n(x)\left(1-e^{-|x-x_0|/\lambda}\right)
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The claim is that the overshoot oscillations are phase-shifted by $\Delta\varphi = \pi$ against the ground state, so they meet the same fate as every other non-optimal branch (e^{iπ} + 1 = 0), and the result is C∞ with 0.00% overshoot.

6.3 The demonstration

The canvas plots a dashed square wave, the classical partial sum in red and the A_0 curve in gold. N sweeps upward from 2, a label marks "+8.95% of jump" on the classical curve, and a freeze button holds N fixed. A small meter compares "classic overshoot ≈ 8.95%" with "A_0 kernel: overshoot → 0".

6.4 The comparison table

Filter kernel	Overshoot	Edge sharpness (rad)	Suppression ratio
Standard Fourier	8.95%	0.02	1.00
Fejér kernel	0.00%	0.45	0.12
Lanczos sigma	1.18%	0.12	0.65
Gottlieb–Shu	0.05%	0.05	0.92
A_0 phase kernel $D^*_n(x)$	0.00%	0.01	1.00

The table's message is that the A_0 kernel is the only entry with zero overshoot, the sharpest edge and full suppression all at once. The other kernels trade one quantity for another: Fejér removes the overshoot but blurs the edge (0.45 rad), and Lanczos keeps 1.18% overshoot. The page gives no derivation for the A_0 row; its numbers follow from the postulated phase annihilation.

7. Applications: structural therapeutics and the 5D monolith

The last two chapters move from theory to engineering: a drug-design method built on distance matrices, and a bronze sculpture presented as an instance of the theory.

7.1 Biomedical structural therapeutics

A biomolecule's 3D coordinates are encoded in a Gram matrix $G_{ij} = r_i \cdot r_j$, and experimental noise pushes its rank above 3. The page proposes recovering the exact conformation by nuclear-norm minimization under a Cayley–Menger constraint:

$$\min_G \|G\|_* \ \text{s.t.}\ \det(\mathrm{CM}_5(G))=0,\ P_\Omega(G)=P_\Omega(G^{\mathrm{exp}})\ \Rightarrow\ \mathrm{rank}(G^*)=3$$

Three application sketches follow.

- **Oncology.** Identification of CM_5 phase cavities on mutant kinases, with small molecules designed to undergo "δ-bond annihilation" only in the mutant binding site and so spare healthy tissue.

- **ALS.** The growth face of TDP-43 and SOD1 fibrils is modelled as a one-dimensional phase crystal; C_2-symmetric peptide inhibitors put a nodal firewall on the growth front and stop monomers from attaching.
- **Parkinson's disease.** R_N-weighted matrix completion designs molecular locks that hold the α-synuclein monomer in a non-aggregating conformation.

The canvas draws a protein backbone with two toggles. With noise on and the constraint off, the page reports rank(G) = 5 and a visibly wobbling, distorted chain. With the constraint on, rank(G) = 3 and the chain is smooth.

7.2 The 5D monolith

The monograph's physical instance is a 300 mm bronze monolith in CuAl10Ni5Fe4, 49.12 kg finished from a 204.66 kg block. The geometric claim is that a 5D penteract (32 vertices) collapses into a 3D cube (8 vertices) under a reduction operator with parameter γ. At γ = 0 the collapse is clean, described as symmetry rotations, phase folding and coalesced field energy density. The slider moves γ from 0 to 1: at 0 the canvas shows an 8-vertex cube in teal, and as γ grows the extra dimensions unfold into the full 32-vertex penteract in gold.

The appendix adds that the dissertation PDF, the STEP and STL models of the sculpture and all source scripts are published openly. The page does not say how a CNC machine would put "field energy density" into the metal. The sculpture is offered as an embodiment of the geometry, not as a test of it.

8. Critical evaluation

As an axiom system the monograph can be read consistently. Its bridges to standard mathematics and physics break at the first elementary check, though, and its central object, the seed A_0, is not what its defining limit says. In this chapter I use standard results as the yardstick, as the page's own honesty clauses invite.

8.1 Summary of checks

Claim on the page	Standard check	Verdict
$A_0 = \lim_{x \to 0} 0/x = \lambda > 0$	$0/x = 0$ for every $x \neq 0$, so the limit is 0; transfer keeps this in $*\mathbb{R}$	Fails

Claim on the page	Standard check	Verdict
A_0 is a fixed point of so(N), yet all structure is its orbit	A fixed point has a one-point orbit; a nonzero vector in $\mathbb{R}^N$ ($N \geq 2$) is not fixed by SO(N), and $UA_0U^\dagger$ is not defined for a vector	Fails as stated
Search in $O(N^2)$ via R_N planes	N^2 counts variables, not input size; a formula can have far more than N^2 clauses; the cube $\{\pm1\}^N$ already lies on a sphere, so the embedding is trivial	Unsupported
$\det(CM_5) = 0$ forbids Navier–Stokes blow-up	Any five points in $\mathbb{R}^3$ have zero 4-volume, so this holds identically	Tautology
$\mathrm{Re}(s) = ½$ forced by $\xi(s) = \xi(1-s)$	The symmetry only makes off-line zeros come in quadruples ρ, $1-\rho$, $\bar{\rho}$, $1-\bar{\rho}$	Not a proof
Mass gap ΔM from vacuum energy $E_0 = \lambda c^2$	The Yang–Mills gap is the spectral gap $E_1 - E_0$, not E_0	Category error
A_0 kernel removes Gibbs overshoot	Multiplying by $1 - e^{-\lvert x-x_0\rvert/\lambda}$ changes the limit function and has a kink at x_0	Fails
A_0 lock removes three-body chaos	Newtonian flow is Hamiltonian, so it has no attractor; locking needs added non-Hamiltonian terms	Contradicts Newton
Every state update costs $\geq k_B T \ln 2$	Landauer's bound applies to logically irreversible erasure only	Overstated
$Q_{diss} = 0$ (Postulate V) and $\Delta E \geq k_B T \ln 2$ (VIII)	The two postulates conflict; a Newtonian universe also has no invariant c (Axiom 3)	Inconsistent
Double-slit: particle plus vacuum field	$I = \cos^2(k\Delta r/2)$ is the standard result; no prediction separates it from other readings	Fair but untested
Perelman retro-read (Poincaré)	Page itself says the mechanism is re-derived, not the credit	Fair

8.2 The seed

Everything depends on A_0, and its definition does not work. The expression 0/x is identically 0 for $x \neq 0$, so the limit is 0. The hyperreal setting does not rescue it: Łoś's transfer principle carries the first-order fact "for all $x \neq 0$, $0/x = 0$" into $*\mathbb{R}$, so $0/\varepsilon = 0$ for every nonzero infinitesimal ε as well. A value $\lambda > 0$ can only enter by definition, not by the limit. The mass-gap reading ($E_0 = \lambda c^2 > 0$) needs exactly that positivity, so it inherits the hole in the argument.

The seed has a second problem. The page calls A_0 the unique fixed point of the full symmetry group, and it also defines the states as the orbit $\{ U A_0 U^\dagger \}$. A fixed point has a one-point orbit, so "every structure is one orbit" would describe a single point. If A_0 is instead a nonzero vector in $\mathbb{R}^N$, as the notation $\lambda \in *\mathbb{R}^N$ suggests, then no such fixed point exists for $N \geqslant 2$, because SO(N) rotates every nonzero vector, and the conjugation $U A_0 U^\dagger$ is not even defined for a vector.

The FAQ defends the division by zero ("we divided by x while x was on its way to zero"). That is a correct statement about limits, and it is exactly why the result is 0.

8.3 P = NP

The honesty clause ("conditional on the Ω_0 convergence layer") is fair, but it hides the whole problem. I see three difficulties.

1. **Cost model.** N^2 grows with the number of variables, but the size of a Boolean formula is set by its clauses, and there can be more than N^2 of them. A procedure that cannot even read its input cannot run in $O(N^2)$.
2. **The embedding is trivial.** The vertices of {−1,+1}^N already lie on a sphere of radius $\sqrt{N}$, so the injection into S^{N−1} adds nothing. Relaxing the cube to a sphere or a rotation group is the standard move behind semidefinite relaxations, and those yield approximations plus a rounding step, not exact solutions.
3. **Convergence is the whole content.** Asking a continuous flow to land on an integer vertex that satisfies the formula is asking it to solve the NP-hard problem. Local minima and saddle points are not addressed.

The page's own Grade C label already says as much. The stronger sentences ("P = NP in $O(N^2)$", "the exponential decision tree does not have to be searched") go beyond it.

8.4 The other Millennium readings

The grades are honest, but several of the chains contain errors that can be checked, beyond mere analogy.

- **Navier–Stokes.** The five-point Cayley–Menger determinant vanishes for any five points in three-dimensional space, so $\det(CM_5) = 0$ cannot single out smooth flows. The notation $\|u\|$ in C^∞ is not a norm, and the Beale–Kato–Majda criterion concerns the time integral of the supremum norm of vorticity, not enstrophy.
- **Riemann.** The functional equation together with conjugation symmetry puts the zeros in sets that are symmetric about the critical line and the real axis. It does not move them onto the line. The canvas draws evenly spaced zeros, whereas the real ones are irregularly spaced.
- **Yang–Mills.** The Clay problem asks for a positive spectral gap above the vacuum in a constructed quantum theory. A positive vacuum energy is a different quantity, and without gravity a constant shift of the energy zero cannot be observed anyway.
- **BSD and Hodge.** No map is defined from rational generators to rotation planes, or from Hodge classes to subobjects of GUT-Cat, so the stated equalities are assertions.

8.5 The Fourier chapter

This is the most testable chapter, and the A_0 kernel fails the test. Multiplying the partial sum by $1 - e^{-|x-x_0|/\lambda}$ is a pointwise envelope applied to the output, not a change of kernel. It pulls the curve toward 0 around the jump and leaves an $O(1)$ error in a neighbourhood of width about λ: at $|x-x_0| = \lambda$ the factor is 0.63, so the curve sits 37% below its target. As N grows the limit is $f(x)(1 - e^{-|x-x_0|/\lambda})$, not f. The overshoot disappears only because the envelope flattens everything within about λ of the jump, which is smoothing by another name. The kink at x_0 also rules out C^∞. The table's "edge sharpness 0.01 rad" is the opposite of what the gold curve shows, a smeared dip. Positive kernels such as Fejér's give zero overshoot at the price of smoothing, which the table records honestly; the A_0 row claims to escape that trade-off without a derivation.

8.6 Physics

- **Double slit.** A classical particle guided by a field that carries the phase is structurally a pilot-wave picture, and pilot-wave theories do reproduce interference. The page gives no field equation beyond $I = \cos^2(k\Delta r/2)$, the ordinary equal-amplitude two-source result, so the demonstration cannot tell this reading apart from standard quantum

mechanics. "Observer-free" also sits awkwardly beside an observer toggle that destroys the pattern. The physical mechanism is which-path information, which the page restates but does not derive.

- **Three-body.** The figure-eight is a genuine periodic solution for equal masses and is linearly stable (shown numerically). But a Hamiltonian flow conserves phase-space volume and cannot have an attractor, so general trajectories cannot be "locked" onto it without non-Newtonian forcing. The twin separation in the demo is scripted to grow at a fixed exponential rate and reset; it is not an integration of the equation shown above it.
- **Time machine.** Reversing the orientation of a trajectory leaves energy unchanged in ordinary mechanics too, so the central statement is correct but not new. Landauer's bound applies only when information is erased, and reversible computation avoids it, so "reversal drives Q to infinity" does not follow.
- **Therapeutics.** A Cayley–Menger constraint on a Gram matrix is a rank condition that distance-geometry methods already use. "δ-bond annihilation" and the three disease claims come with no binding data on the page.

8.7 Internal consistency

Beyond the table, I see three tensions. The FAQ says that, inside Ω_0, the seven readings are "theorems of the axiom system", while five of the cards carry the label "structural analogy"; the page does not say how the two labels relate. The theory is called "special" because its computational metric is fixed and flat, yet Postulate V (a "perfect Newtonian universe" with zero dissipation) and Axiom 3 (an invariant speed c) belong to different physical frameworks. And the FAQ contains jokes (a cancer cure, "Give me a bit of time", and the mass gap between the reader and the fridge) that blur the line between claim and humour, so a reader cannot always tell which statements are meant to be tested.

8.8 Falsifiability

The page offers three tests. (1) The Millennium readings are mathematical claims, so a single error in an embedding brings them down; sections 8.2 to 8.5 point to such errors. (2) The firewall reading of interference is called testable "against any double-slit configuration", but it reproduces the standard intensity, so no experiment is proposed that could separate it from ordinary quantum mechanics. (3) The claim that no causal loop survives a firewall with $|\psi| = 0$ is true by construction, since $|\psi| = 0$ is postulated. Only the first test has any force.

8.9 What survives

What I like about the page is its grading, its honesty clauses, and the way it turns abstract statements into controllable animations that expose their own assumptions: switch a postulate off and watch what changes. The plane count $R_N = N(N-1)/2$, the Gibbs constant and the figure-eight are all stated correctly. What does not survive is the step from a shared notation (phase, rotation, anti-phase annihilation) to a shared mechanism.

9. Conclusion

The Special Theory of Computational Relativity is best read as a notation and a set of illustrations built on one unproved postulate, not as a resolution of the problems it names. The rotation-plane count $R_N = N(N-1)/2$, the Gibbs constant of 8.95% and the figure-eight orbit are correct. The seed A_0, the $O(N^2)$ search, the Fourier kernel and the Navier–Stokes argument do not survive elementary checks.

9.1 Findings

1. The theory is a consistent axiom system only if the seed is defined by fiat as a nonzero element; the limit that is supposed to define it equals 0.
2. The author grades five of the seven Millennium readings as analogies, and the evaluation supports that grade. The P = NP reading depends on a convergence step that is the whole difficulty.
3. The strongest quantitative claim, the zero-overshoot A_0 kernel, is testable and fails: the construction changes the limit function.
4. The page is open about its status in many places, and its interactive format lets a reader inspect the assumptions. That is a real strength.

9.2 Repairs that would make the theory testable

- Define λ as an axiom (a nonzero element of $*\mathbb{R}$) and drop the limit, or change the seed.
- State the P = NP claim as a conjecture about one specific flow, with a proof or a numerical benchmark of convergence on random 3-SAT.
- Replace the A_0 envelope with a derived kernel, and report overshoot, edge width and L^2 error against Fejér, Lanczos and Gottlieb–Shu on the same square wave.
- Give the vacuum field in the double-slit reading a dynamical equation, and state one prediction that differs from standard quantum mechanics.

- Resolve the conflict between Postulates V and VIII, and restrict Landauer's bound to erasure.
- Reconcile the FAQ with the grades: say which readings are theorems inside Ω_0 and which remain analogies.

9.3 Future work named by the page

The page leaves open a general theory in which the computational metric curves under problem load, by analogy with the step from special to general relativity. Nothing in this text constrains what that metric would be, so the sensible next step is to state definitions before drawing conclusions.

Appendix: axioms, postulates and invariants

The table reproduces the page's own appendix. The page distinguishes axioms (the unprovable cornerstones) from postulates (the physical and algorithmic laws of dynamics).

Label	Type / name	Key expression
Axiom 1	The seed A_0 (zero-point singularity)	$A_0 \equiv \lim(x\to0)\ 0/x = \lambda \in *\mathbb{R}^N$
Axiom 2	Hyperreal topos, Łoś transfer	$*\mathbb{V} = V^I/$
Axiom 3	Invariance of c and rest energy E_0	$E_0 = \int u\, d^3r = mc^2$
Postulate I	Euler anti-phase annihilation	$e^{i\pi} + 1 = 0 \Rightarrow \lvert\psi_node\rvert = 0$
Postulate II	so(N) embedding of complexity	$R_N = N(N-1)/2$ planes, $O(N^2)$
Postulate III	Cayley–Menger volume invariance	$\det(CM_5) = 0$ in 3D
Postulate IV	Multi-lobe orbital hierarchy	δ, φ, octadic lattice $_8$
Postulate V	Zero entropic friction ("Perfect Newtonian Universe")	$Q_diss = 0 \Rightarrow t_alg = t_phys$
Postulate VI	Dirichlet phase filtering (A_0 kernel)	$D^*_n(x) = D_n(x)(1 - e^{-\lvert x-x_0\rvert/\lambda})$

Label	Type / name	Key expression
Postulate VII	Causal sector isolation	$\|\psi(\Sigma_firewall)\|^2 = 0$
Postulate VIII	Landauer thermodynamic bound	$\Delta E \geqslant k_BT \ln 2$

Sources

- *The Special Theory of Computational Relativity*, CoReX v3.1, interactive monograph by Pierre LeBaux (the page this dissertation is based on, in the corrected build corex-v3-4.html). All quotations, equations, numbers and demonstrations come from it.
- Standard results used only in Chapter 8 (limits and transfer, Cayley–Menger determinants, the Beale–Kato–Majda criterion, Landauer's bound, Hamiltonian phase-space conservation, the Chenciner–Montgomery figure-eight) are cited from general knowledge, not looked up for this text.